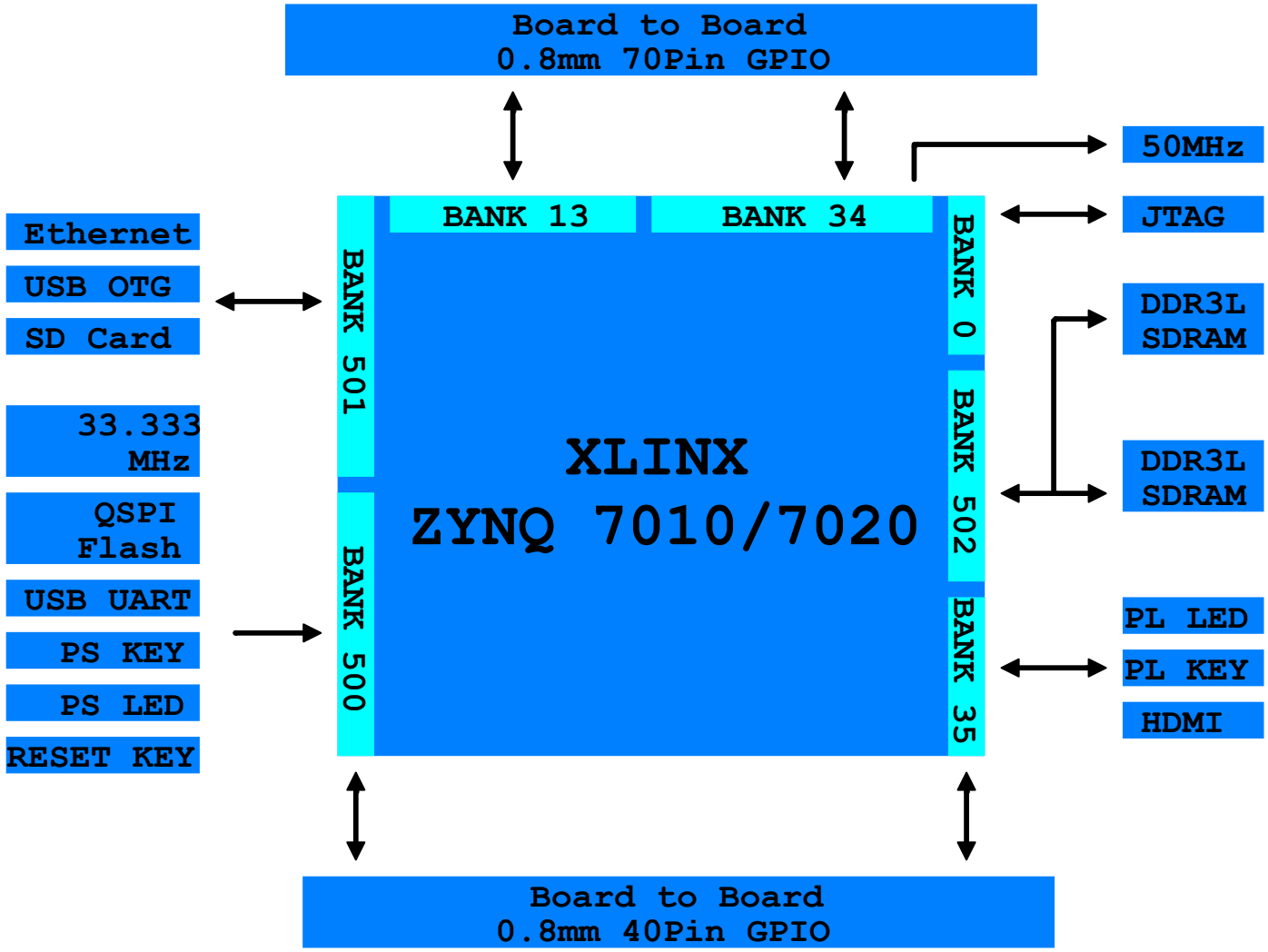
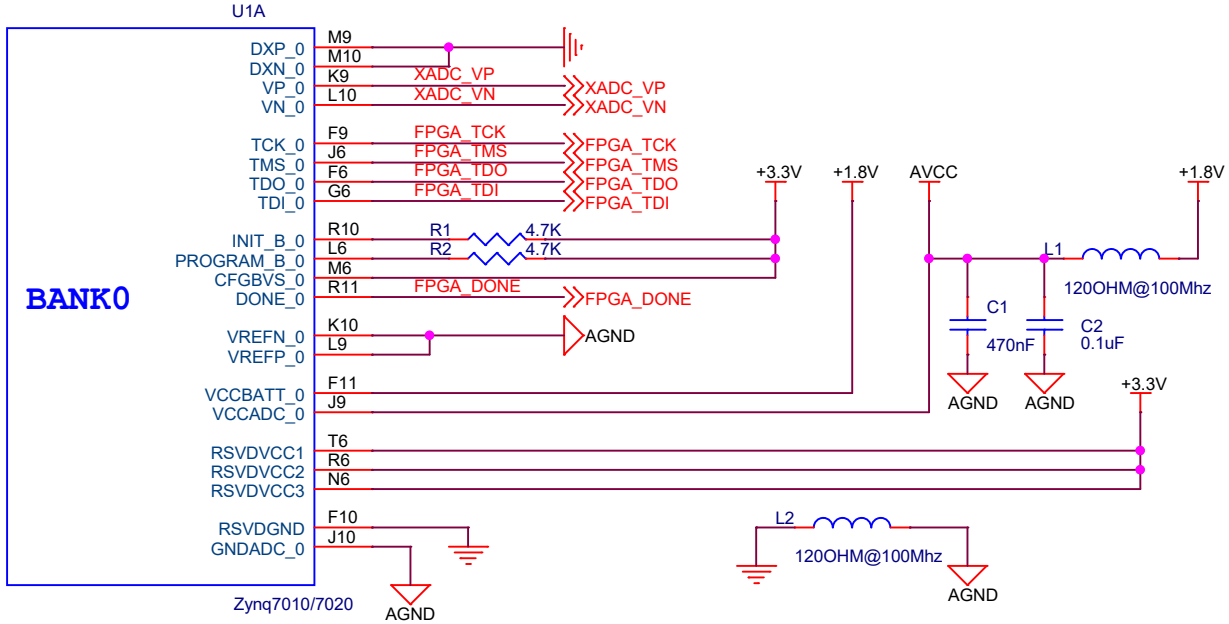
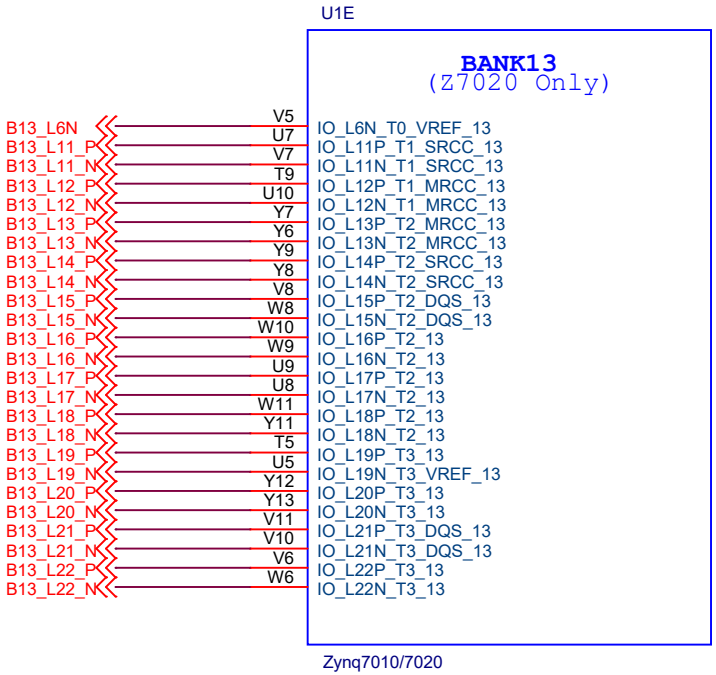


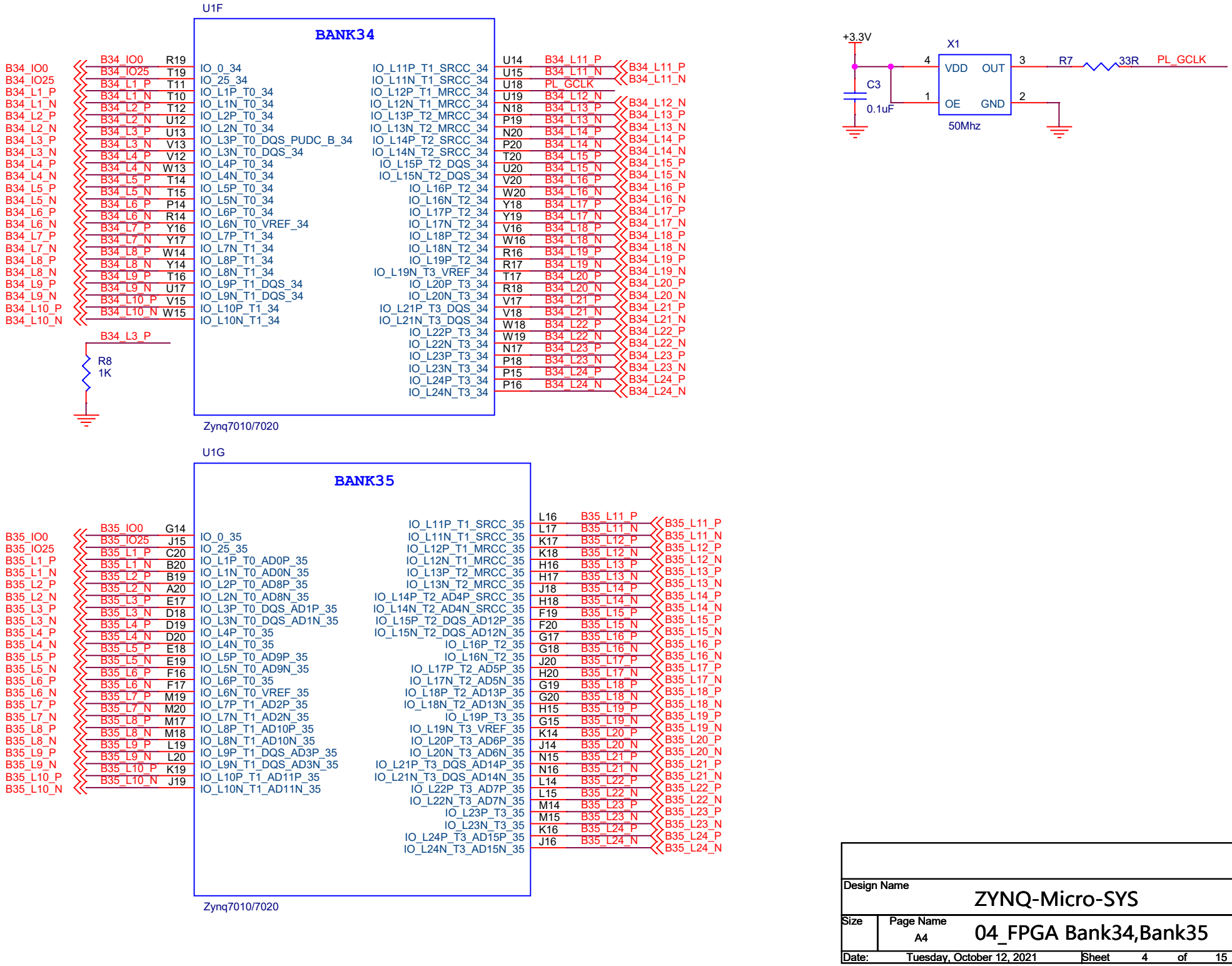
Design Name	ZYNQ-Micro-SYS	
Size: A4	Page Name: 01_Block Diagram	Rev: 1.2
Date: Tuesday, October 12, 2021	Sheet 1 of 15	

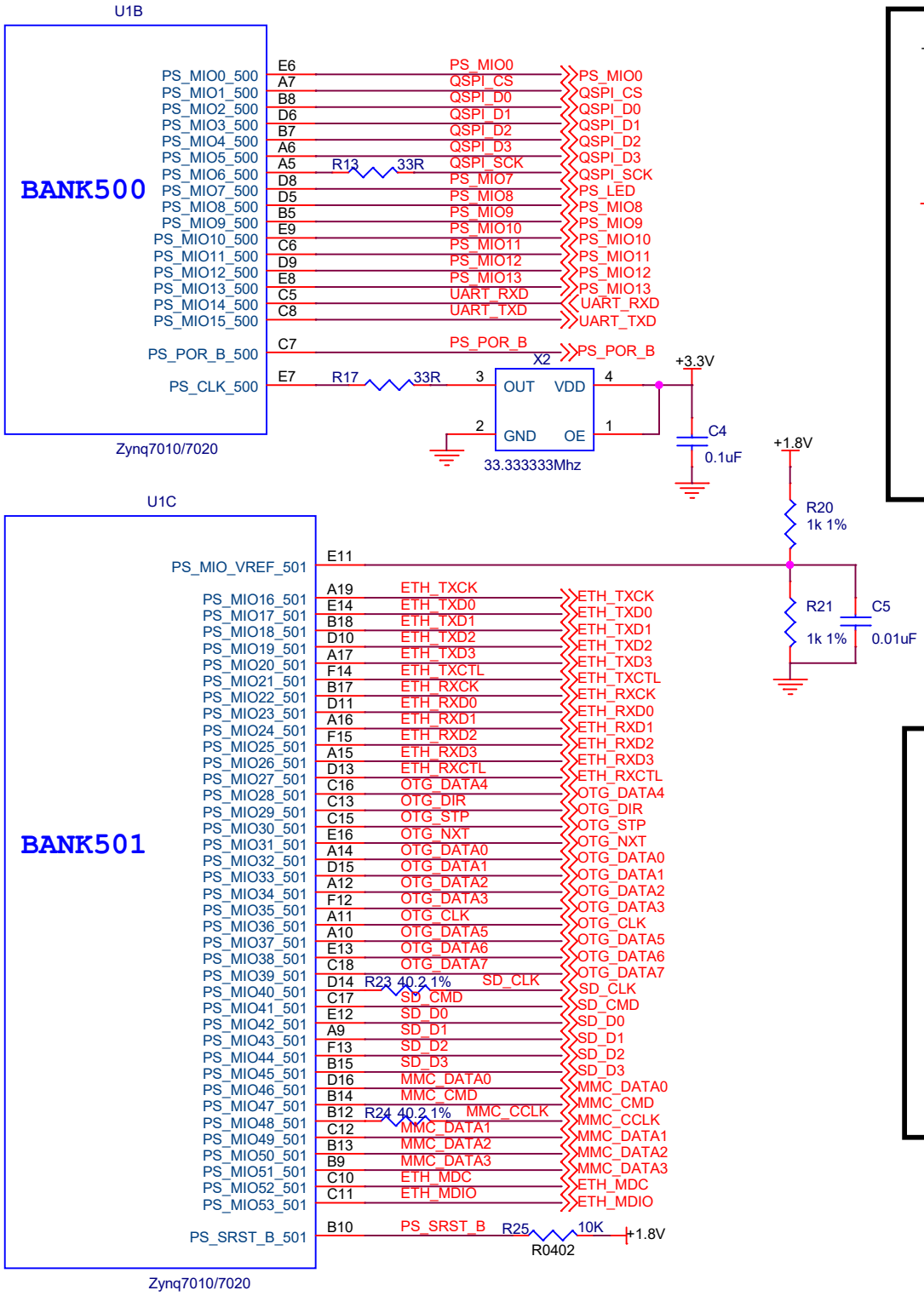


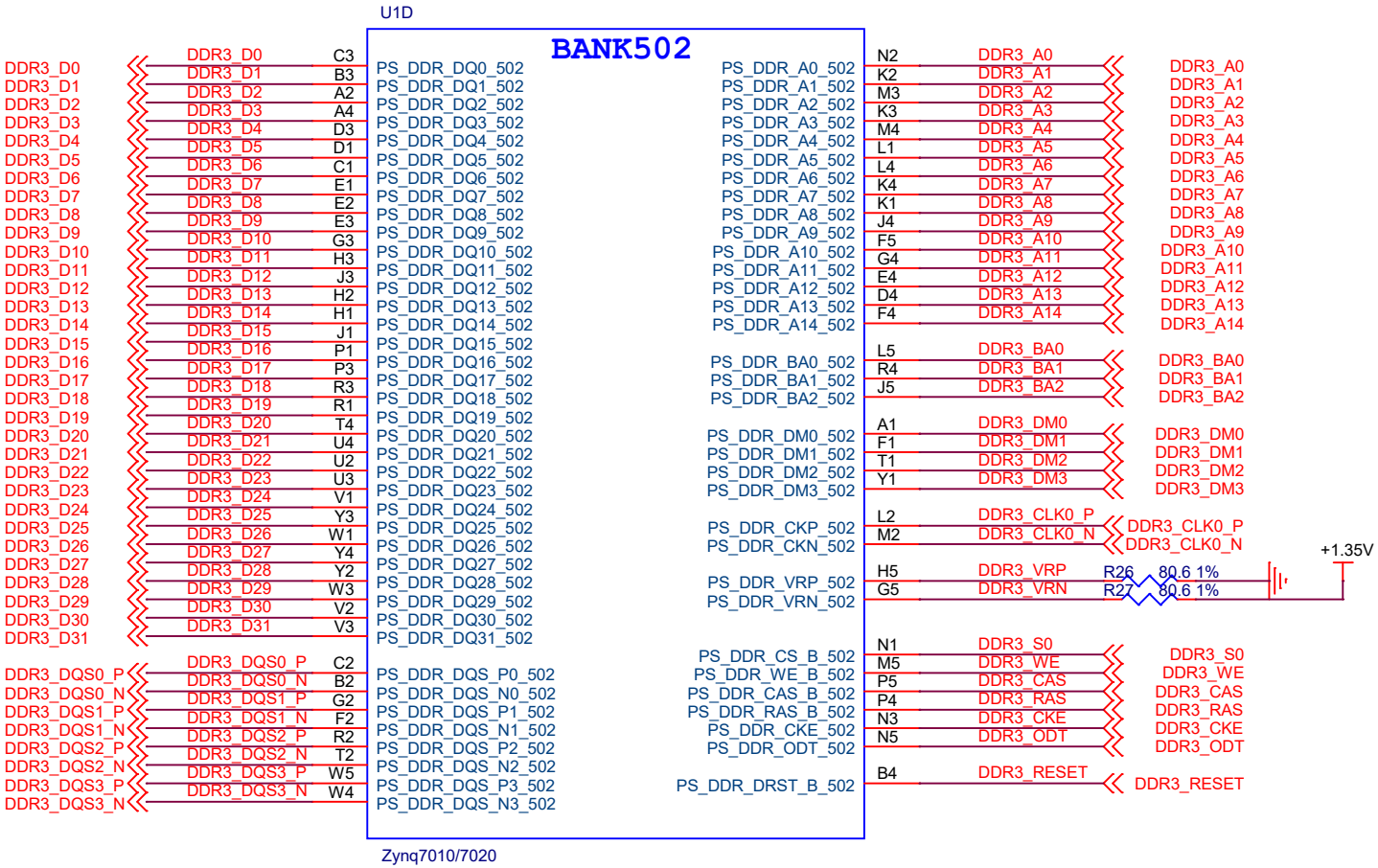
Design Name	ZYNQ-Micro-SYS		
Size	Page Name A4	02_FPGA Bank0	Rev 1.2
Date: Tuesday, October 12, 2021	Sheet 2 of 15		



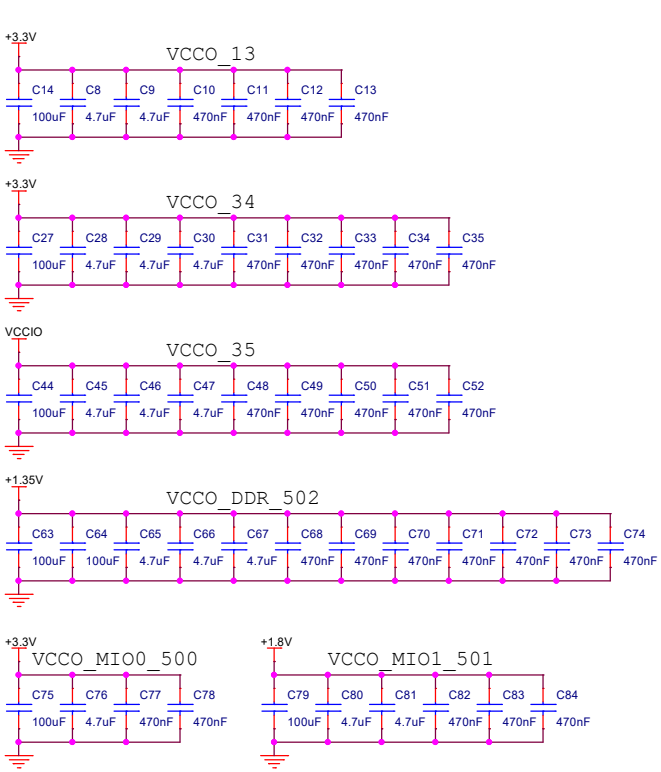
Design Name	ZYNQ-Micro-SYS	
Size	Page Name: A4 — 03_FPGA Bank_13	Rev: 1.2
Date: Tuesday, October 12, 2021	Sheet 3 of 15	

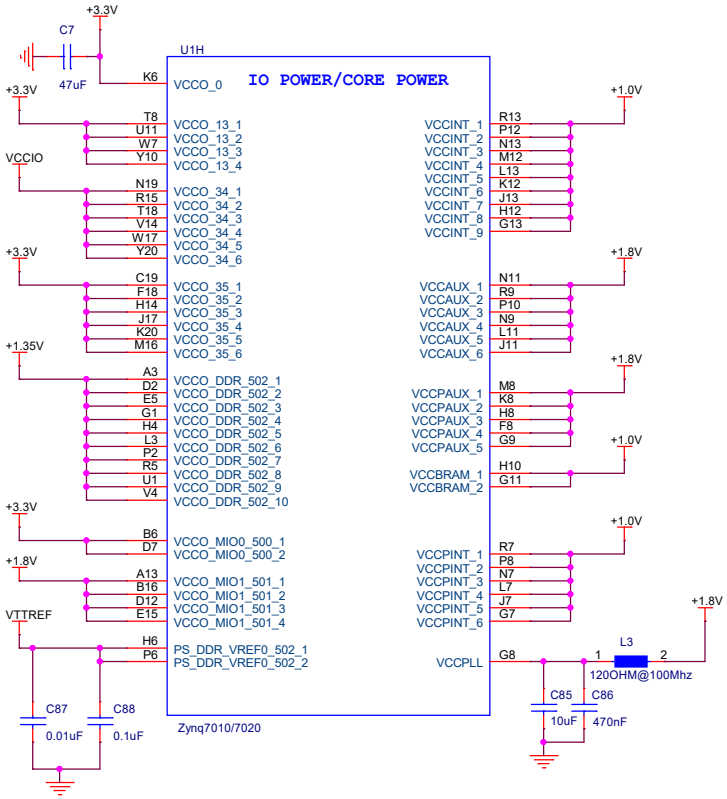

U1F
BANK34
B34_IO0 R19
B34_IO25 T19
B34_L1_P T11
B34_L1_N T10
B34_L2_P T12
B34_L2_N U12
B34_L3_P U13
B34_L3_N V13
B34_L4_P V12
B34_L4_N W13
B34_L5_P T14
B34_L5_N T15
B34_L6_P P14
B34_L6_N R14
B34_L7_P Y16
B34_L7_N Y17
B34_L8_P W14
B34_L8_N Y14
B34_L9_P T16
B34_L9_N U17
B34_L10_P V15
B34_L10_N W15
IO_0_34
IO_25_34
IO_L1P_T0_34
IO_L1N_T0_34
IO_L2P_T0_34
IO_L2N_T0_34
IO_L3P_T0_DQS_PUDC_B_34
IO_L3N_T0_DQS_34
IO_L4P_T0_34
IO_L4N_T0_34
IO_L5P_T0_34
IO_L5N_T0_34
IO_L6P_T0_34
IO_L6N_T0_VREF_34
IO_L7P_T1_34
IO_L7N_T1_34
IO_L8P_T1_34
IO_L8N_T1_34
IO_L9P_T1_DQS_34
IO_L9N_T1_DQS_34
IO_L10P_T1_34
IO_L10N_T1_34
IO_L11P_T1_SRCC_34 U14 B34_L11_P
IO_L11N_T1_SRCC_34 U15 B34_L11_N
IO_L12P_T1_MRCC_34 U18 PL_GCLK
IO_L12N_T1_MRCC_34 U19 B34_L12_N
IO_L13P_T2_MRCC_34 N18 B34_L13_P
IO_L13N_T2_MRCC_34 P19 B34_L13_N
IO_L14P_T2_SRCC_34 N20 B34_L14_P
IO_L14N_T2_SRCC_34 P20 B34_L14_N
IO_L15P_T2_DQS_34 T20 B34_L15_P
IO_L15N_T2_DQS_34 U20 B34_L15_N
IO_L16P_T2_34 V20 B34_L16_P
IO_L16N_T2_34 W20 B34_L16_N
IO_L17P_T2_34 Y18 B34_L17_P
IO_L17N_T2_34 Y19 B34_L17_N
IO_L18P_T2_34 V16 B34_L18_P
IO_L18N_T2_34 W16 B34_L18_N
IO_L19P_T2_34 R16 B34_L19_P
IO_L19N_T3_VREF_34 R17 B34_L19_N
IO_L20P_T3_34 T17 B34_L20_P
IO_L20N_T3_34 R18 B34_L20_N
IO_L21P_T3_DQS_34 V17 B34_L21_P
IO_L21N_T3_DQS_34 V18 B34_L21_N
IO_L22P_T3_34 W18 B34_L22_P
IO_L22N_T3_34 W19 B34_L22_N
IO_L23P_T3_34 N17 B34_L23_P
IO_L23N_T3_34 P18 B34_L23_N
IO_L24P_T3_34 P15 B34_L24_P
IO_L24N_T3_34 P16 B34_L24_N
B34_L3_P
R8
1K
Zynq7010/7020
U1G
BANK35
B35_IO0 G14
B35_IO25 J15
B35_L1_P C20
B35_L1_N B20
B35_L2_P B19
B35_L2_N A20
B35_L3_P E17
B35_L3_N D18
B35_L4_P D19
B35_L4_N D20
B35_L5_P E18
B35_L5_N E19
B35_L6_P F16
B35_L6_N F17
B35_L7_P M19
B35_L7_N M20
B35_L8_P M17
B35_L8_N M18
B35_L9_P L19
B35_L9_N L20
B35_L10_P K19
B35_L10_N J19
IO_0_35
IO_25_35
IO_L1P_T0_AD0P_35
IO_L1N_T0_AD0N_35
IO_L2P_T0_AD8P_35
IO_L2N_T0_AD8N_35
IO_L3P_T0_DQS_AD1P_35
IO_L3N_T0_DQS_AD1N_35
IO_L4P_T0_35
IO_L4N_T0_35
IO_L5P_T0_AD9P_35
IO_L5N_T0_AD9N_35
IO_L6P_T0_35
IO_L6N_T0_VREF_35
IO_L7P_T1_AD2P_35
IO_L7N_T1_AD2N_35
IO_L8P_T1_AD10P_35
IO_L8N_T1_AD10N_35
IO_L9P_T1_DQS_AD3P_35
IO_L9N_T1_DQS_AD3N_35
IO_L10P_T1_AD11P_35
IO_L10N_T1_AD11N_35
IO_L11P_T1_SRCC_35 L16 B35_L11_P
IO_L11N_T1_SRCC_35 L17 B35_L11_N
IO_L12P_T1_MRCC_35 K17 B35_L12_P
IO_L12N_T1_MRCC_35 K18 B35_L12_N
IO_L13P_T2_MRCC_35 H16 B35_L13_P
IO_L13N_T2_MRCC_35 H17 B35_L13_N
IO_L14P_T2_AD4P_SRCC_35 J18 B35_L14_P
IO_L14N_T2_AD4N_SRCC_35 H18 B35_L14_N
IO_L15P_T2_DQS_AD12P_35 F19 B35_L15_P
IO_L15N_T2_DQS_AD12N_35 F20 B35_L15_N
IO_L16P_T2_35 G17 B35_L16_P
IO_L16N_T2_35 G18 B35_L16_N
IO_L17P_T2_AD5P_35 J20 B35_L17_P
IO_L17N_T2_AD5N_35 H20 B35_L17_N
IO_L18P_T2_AD13P_35 G19 B35_L18_P
IO_L18N_T2_AD13N_35 G20 B35_L18_N
IO_L19P_T3_35 H15 B35_L19_P
IO_L19N_T3_VREF_35 G15 B35_L19_N
IO_L20P_T3_AD6P_35 K14 B35_L20_P
IO_L20N_T3_AD6N_35 J14 B35_L20_N
IO_L21P_T3_DQS_AD14P_35 N15 B35_L21_P
IO_L21N_T3_DQS_AD14N_35 N16 B35_L21_N
IO_L22P_T3_AD7P_35 L14 B35_L22_P
IO_L22N_T3_AD7N_35 L15 B35_L22_N
IO_L23P_T3_35 M14 B35_L23_P
IO_L23N_T3_35 M15 B35_L23_N
IO_L24P_T3_AD15P_35 K16 B35_L24_P
IO_L24N_T3_AD15N_35 J16 B35_L24_N
Zynq7010/7020
+3.3V
X1
4
VDD
OUT
3
R7
33R
PL_GCLK
C3
0.1uF
1
OE
GND
2
50Mhz
Design Name
ZYNQ-Micro-SYS
Size
Page Name
A4
04_FPGA Bank34,Bank35
Rev
1.2
Date:
Tuesday, October 12, 2021
Sheet
4
of
15

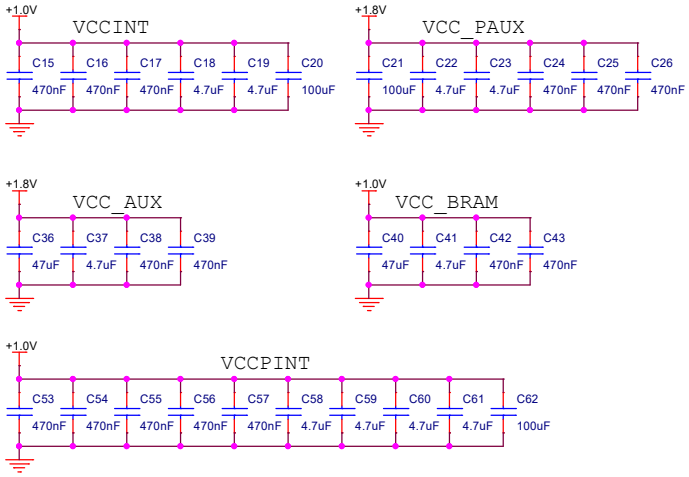

U1B
BANK500
PS_MIO0_500
PS_MIO1_500
PS_MIO2_500
PS_MIO3_500
PS_MIO4_500
PS_MIO5_500
PS_MIO6_500
PS_MIO7_500
PS_MIO8_500
PS_MIO9_500
PS_MIO10_500
PS_MIO11_500
PS_MIO12_500
PS_MIO13_500
PS_MIO14_500
PS_MIO15_500
PS_POR_B_500
PS_CLK_500
E6 PS_MIO0
A7 QSPI_CS
B8 QSPI_D0
D6 QSPI_D1
B7 QSPI_D2
A6 QSPI_D3
A5 QSPI_SCK
D8 PS_MIO7
D5 PS_MIO8
B5 PS_MIO9
E9 PS_MIO10
C6 PS_MIO11
D9 PS_MIO12
E8 PS_MIO13
C5 UART_RXD
C8 UART_TXD
C7 PS_POR_B
E7
R13 33R
R17 33R
X2
OUT VDD
GND OE
33.333333Mhz
+3.3V
C4 0.1uF
Zynq7010/7020
U1C
BANK501
PS_MIO_VREF_501
PS_MIO16_501
PS_MIO17_501
PS_MIO18_501
PS_MIO19_501
PS_MIO20_501
PS_MIO21_501
PS_MIO22_501
PS_MIO23_501
PS_MIO24_501
PS_MIO25_501
PS_MIO26_501
PS_MIO27_501
PS_MIO28_501
PS_MIO29_501
PS_MIO30_501
PS_MIO31_501
PS_MIO32_501
PS_MIO33_501
PS_MIO34_501
PS_MIO35_501
PS_MIO36_501
PS_MIO37_501
PS_MIO38_501
PS_MIO39_501
PS_MIO40_501
PS_MIO41_501
PS_MIO42_501
PS_MIO43_501
PS_MIO44_501
PS_MIO45_501
PS_MIO46_501
PS_MIO47_501
PS_MIO48_501
PS_MIO49_501
PS_MIO50_501
PS_MIO51_501
PS_MIO52_501
PS_MIO53_501
PS_SRST_B_501
E11
A19 ETH_TXCK
E14 ETH_TXD0
B18 ETH_TXD1
D10 ETH_TXD2
A17 ETH_TXD3
F14 ETH_TXCTL
B17 ETH_RXCK
D11 ETH_RXD0
A16 ETH_RXD1
F15 ETH_RXD2
A15 ETH_RXD3
D13 ETH_RXCTL
C16 OTG_DATA4
C13 OTG_DIR
C15 OTG_STP
E16 OTG_NXT
A14 OTG_DATA0
D15 OTG_DATA1
A12 OTG_DATA2
F12 OTG_DATA3
A11 OTG_CLK
A10 OTG_DATA5
E13 OTG_DATA6
C18 OTG_DATA7
D14 R23 40.2 1% SD_CLK
C17 SD_CMD
E12 SD_D0
A9 SD_D1
F13 SD_D2
B15 SD_D3
D16 MMC_DATA0
B14 MMC_CMD
B12 R24 40.2 1% MMC_CCLK
C12 MMC_DATA1
B13 MMC_DATA2
B9 MMC_DATA3
C10 ETH_MDC
C11 ETH_MDIO
B10 PS_SRST_B
R25 10K
R0402
+1.8V
R20 1k 1%
R21 1k 1%
C5 0.01uF
Zynq7010/7020
VOLTAGE/JTAG/PLL OPTION
+3.3V
R9 20K PS_MIO8
R10 20K QSPI_D0
R11 20K QSPI_D1
R12 20K QSPI_SCK
R14 20K PS_MIO7
MIO[8] = 1 ---- MIO bank1 voltage = 1.8V
MIO[2] = 0 ---- cascaded JTAG
MIO[3] = 0 ---- JTAG/NAND/Quad-SPI/SD
MIO[6] = 0 ---- PLL used
MIO[7] = 0 ---- MIO bank0 voltage = 3.3V
POWER ON RESET
KEY1
BUTTON
U2
#MR VDD
GND #RESET
TCM811TERCTR
+3.3V
C6 0.1uF
R22 4.7K
PS_POR_B
Design Name
ZYNQ-Micro-SYS
Size
Page Name
A4
05_FPGA Bank500,Bank501
Rev
1.2
Date: Tuesday, October 12, 2021
Sheet 5 of 15

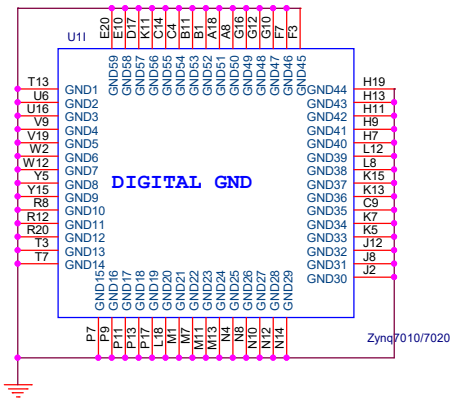


Design Name	ZYNQ-Micro-SYS	
Size	Page Name: A4 — 06_FPGA Bank502	Rev: 1.2
Date: Tuesday, October 12, 2021	Sheet 6 of 15	

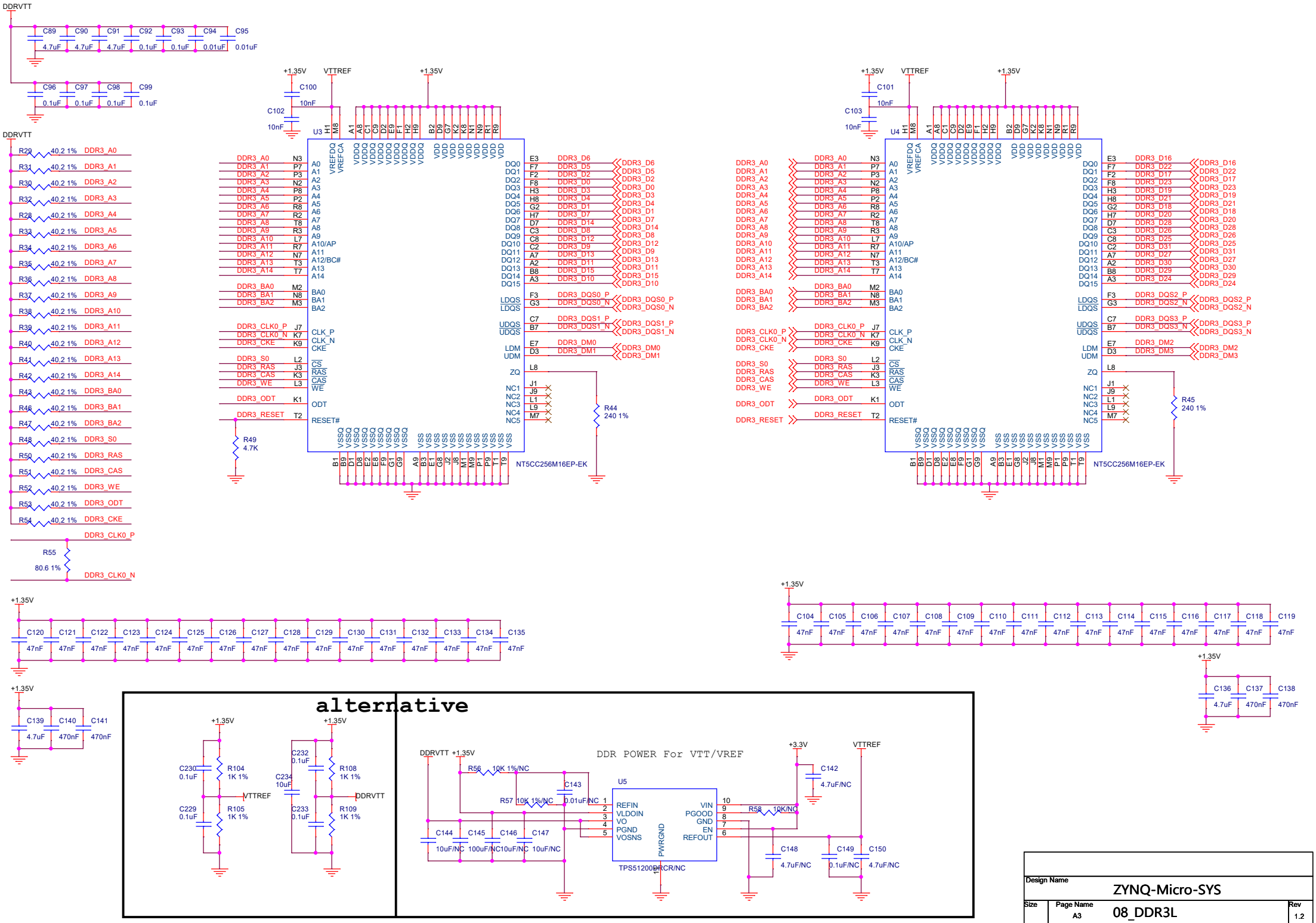


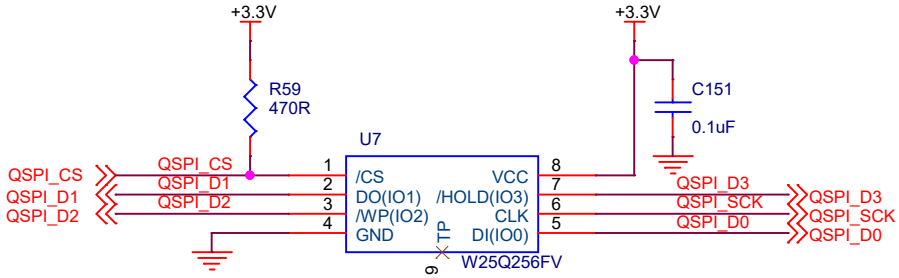






Design Name	ZYNQ-Micro-SYS		
Size	Page Name A3	07_FPGA POWER	Rev 1.2
Date: Tuesday, October 12, 2021	Sheet 7 of 15		

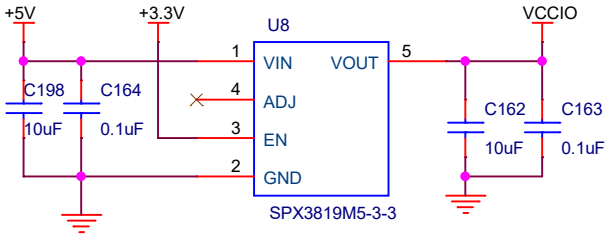

DDRVTT
C89 C90 C91 C92 C93 C94 C95
4.7uF 4.7uF 4.7uF 0.1uF 0.1uF 0.01uF 0.01uF
C96 C97 C98 C99
0.1uF 0.1uF 0.1uF 0.1uF
+1.35V
VTTREF
C100 10nF
C102 10nF
U3
NT5CC256M16EP-EK
R49 4.7K
R44 240 1%
C101 10nF
C103 10nF
U4
R45 240 1%
R29 R31 R30 R32 R28 R33 R34 R35 R36 R37 R38 R39 R40 R41 R42 R43 R46 R47 R48 R50 R51 R52 R53 R54 40.2 1%
DDR3_A0 DDR3_A1 DDR3_A2 DDR3_A3 DDR3_A4 DDR3_A5 DDR3_A6 DDR3_A7 DDR3_A8 DDR3_A9 DDR3_A10 DDR3_A11 DDR3_A12 DDR3_A13 DDR3_A14 DDR3_BA0 DDR3_BA1 DDR3_BA2 DDR3_S0 DDR3_RAS DDR3_CAS DDR3_WE DDR3_ODT DDR3_CKE
DDR3_CLK0_P
R55 80.6 1%
DDR3_CLK0_N
C120 C121 C122 C123 C124 C125 C126 C127 C128 C129 C130 C131 C132 C133 C134 C135 47nF
C104 C105 C106 C107 C108 C109 C110 C111 C112 C113 C114 C115 C116 C117 C118 C119 47nF
C136 4.7uF
C137 470nF
C138 470nF
C139 4.7uF
C140 470nF
C141 470nF
alternative
C230 0.1uF
R104 1K 1%
C229 0.1uF
R105 1K 1%
VTTREF
C232 0.1uF
C234 10uF
R108 1K 1%
C233 0.1uF
R109 1K 1%
DDRVTT
DDR POWER For VTT/VREF
R56 10K 1%/NC
R57 10K 1%/NC
C143 0.01uF/NC
U5
REFIN
VLDOIN
VO
PGND
VOSNS
PWRGND
VIN
PGOOD
GND
EN
REFOUT
R58 10K/NC
C142 4.7uF/NC
C144 10uF/NC
C145 100uF/NC
C146 10uF/NC
C147 10uF/NC
C148 4.7uF/NC
C149 0.1uF/NC
C150 4.7uF/NC
TPS51200PRCR/NC
+3.3V
Design Name
ZYNQ-Micro-SYS
Size
A3
Page Name
08_DDR3L
Rev
1.2
Date:
Tuesday, October 12, 2021
Sheet
8
of
15



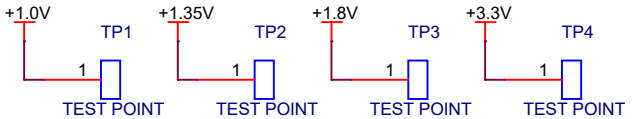
Design Name	ZYNQ-Micro-SYS	
Size	Page Name: A4 — 09_QSPI	Rev: 1.2
Date: Tuesday, October 12, 2021	Sheet 9 of 15	

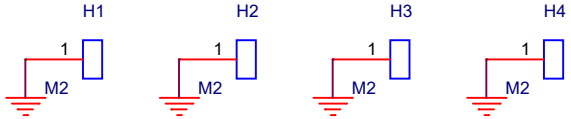
POWER ON: VCCINT(1.0V)->VCCBRAM(1.0V)->VCCAUX-(1.8V)>VCCO(1.5V and 3.3V)

Vout=0.6*R1/R2+0.6



Power On Sequnce:
1.0V -> 1.8V -> 1.35V -> 3.3V -> VCCIO



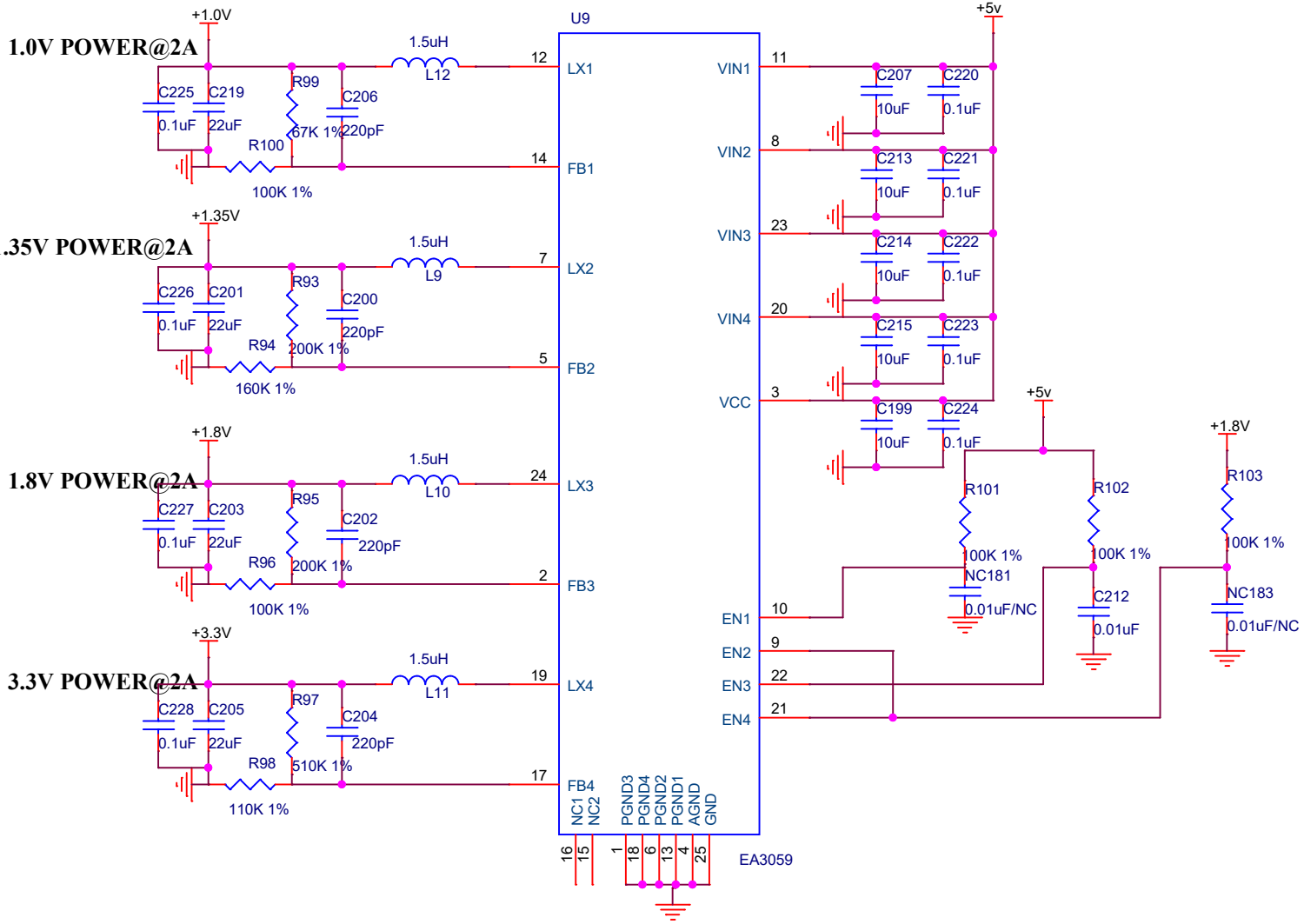


1.0V POWER@2A

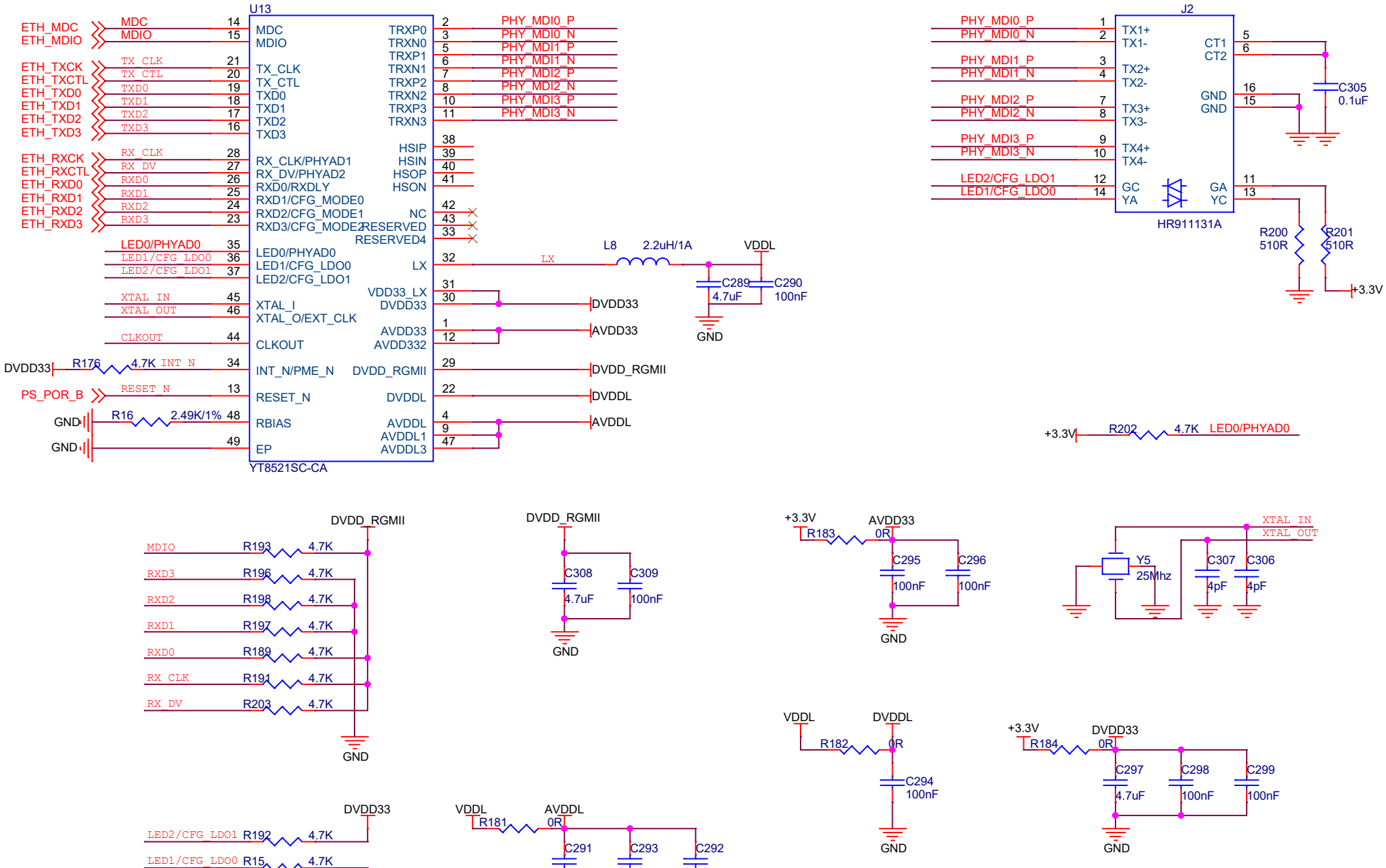
1.35V POWER@2A

1.8V POWER@2A

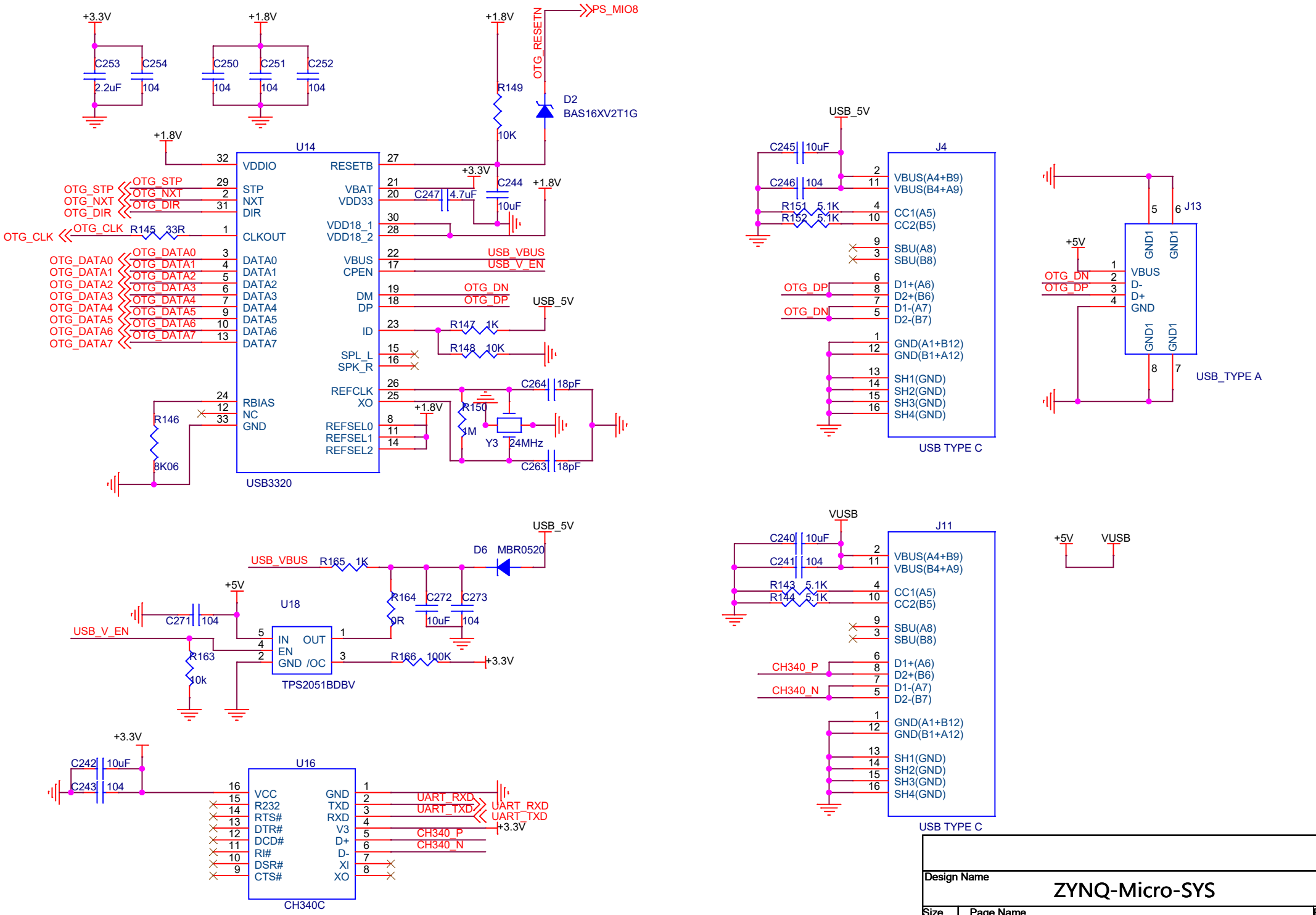
3.3V POWER@2A

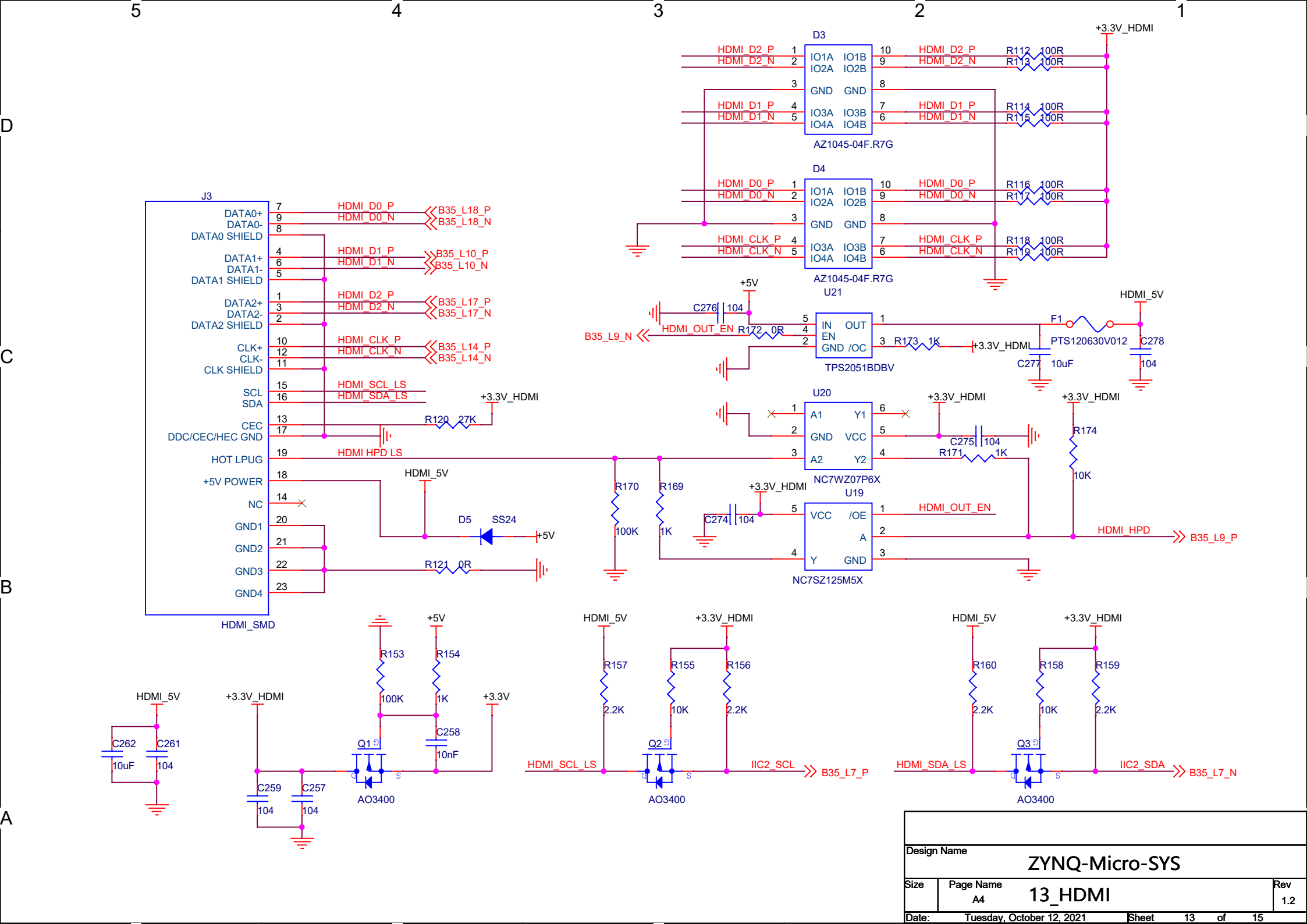


Design Name	ZYNQ-Micro-SYS		
Size	Page Name A4	10_POWER	Rev 1.2
Date: Tuesday, October 12, 2021	Sheet 10 of 15		

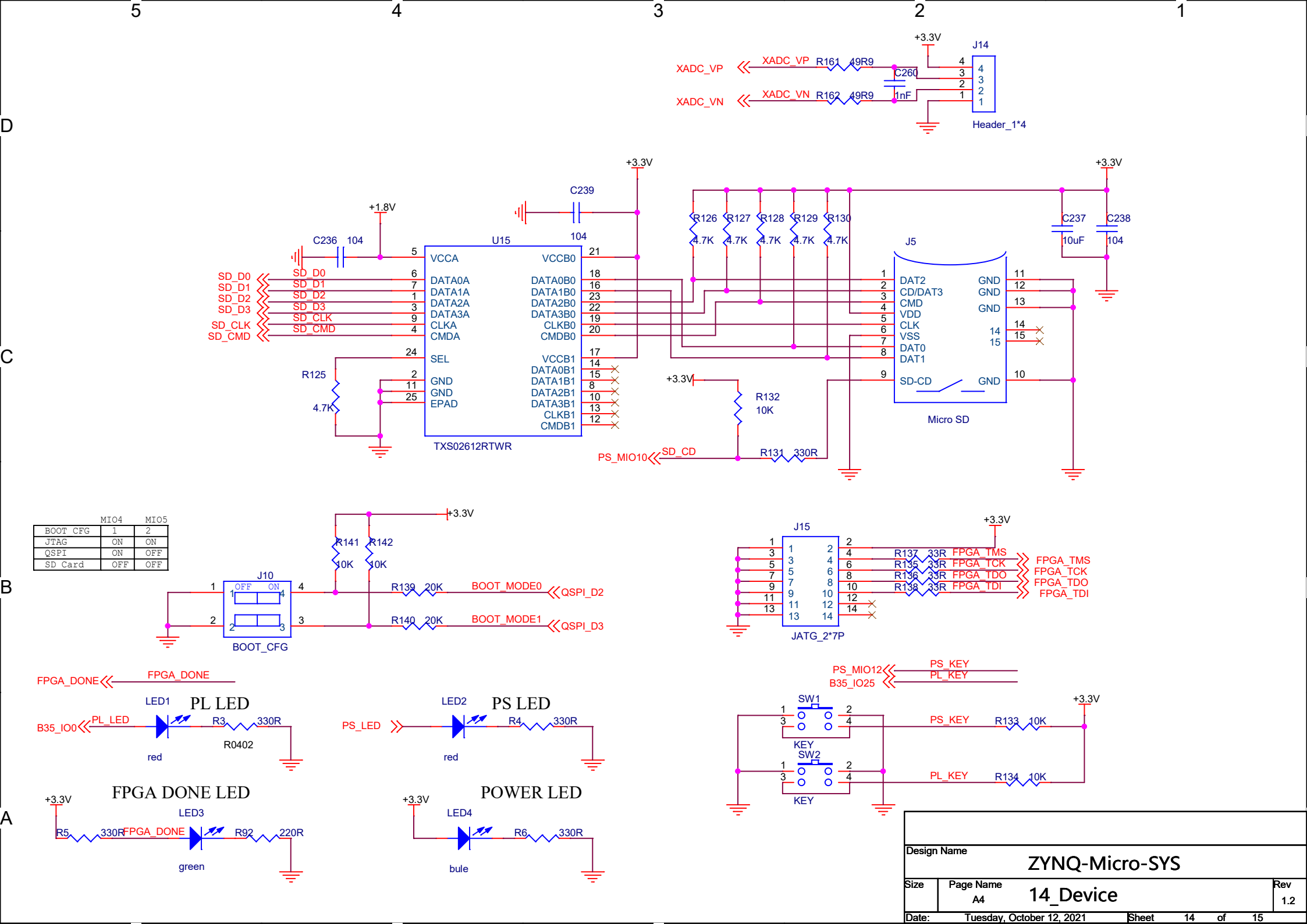


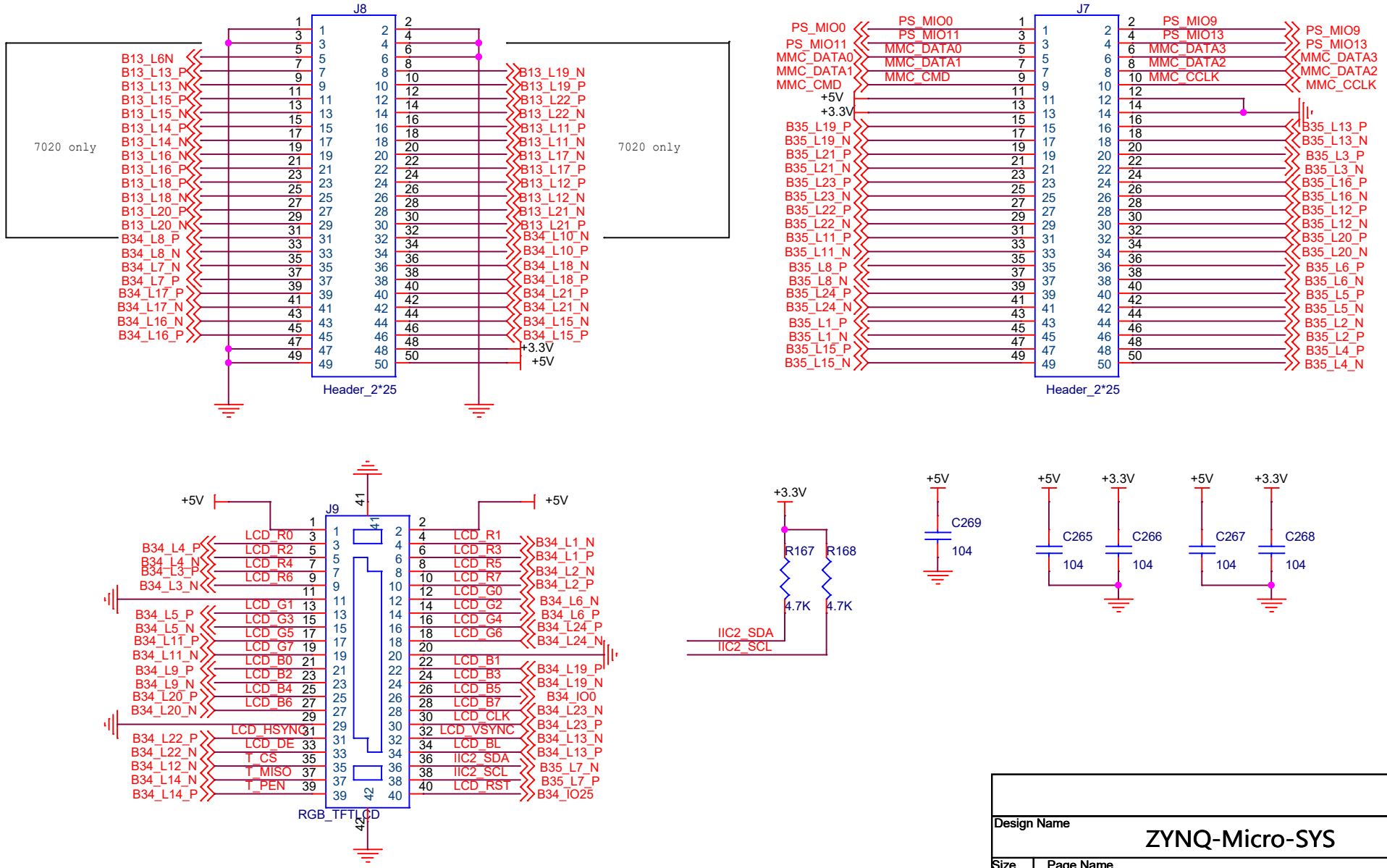
Design Name	ZYNQ-Micro-SYS	
Size	Page Name A4	11_Ethernet
		Rev 1.2
Date: Tuesday, October 12, 2021	Sheet 11 of 15	


5
4
3
2
1
D
C
B
A
+3.3V
+1.8V
C253
C254
2.2uF
104
C250
C251
C252
104
104
104
U14
VDDIO
RESETB
VBAT
VDD33
VDD18_1
VDD18_2
VBUS
CPEN
DM
DP
ID
SPL_L
SPK_R
REFCLK
XO
REFSEL0
REFSEL1
REFSEL2
RBIAS
NC
GND
STP
NXT
DIR
CLKOUT
DATA0
DATA1
DATA2
DATA3
DATA4
DATA5
DATA6
DATA7
OTG_STP
OTG_NXT
OTG_DIR
OTG_CLK
OTG_DATA0
OTG_DATA1
OTG_DATA2
OTG_DATA3
OTG_DATA4
OTG_DATA5
OTG_DATA6
OTG_DATA7
R145 33R
R146
8K06
USB3320
R149
10K
OTG_RESETN
PS_MIO8
D2
BAS16XV2T1G
C247 4.7uF
C244
10uF
USB_VBUS
USB_V_EN
OTG_DN
OTG_DP
USB_5V
R147 1K
R148 10K
C264 18pF
C263 18pF
R150
1M
Y3 24MHz
USB_5V
C245 10uF
C246 104
R151 5.1K
R152 5.1K
J4
VBUS(A4+B9)
VBUS(B4+A9)
CC1(A5)
CC2(B5)
SBU(A8)
SBU(B8)
D1+(A6)
D2+(B6)
D1-(A7)
D2-(B7)
GND(A1+B12)
GND(B1+A12)
SH1(GND)
SH2(GND)
SH3(GND)
SH4(GND)
USB TYPE C
J13
GND1
VBUS
D-
D+
GND
USB_TYPE A
USB_VBUS
R165 1K
D6 MBR0520
U18
IN
OUT
EN
GND /OC
TPS2051BDBV
C271 104
R163 10k
R164 0R
C272 10uF
C273 104
R166 100K
VUSB
C240 10uF
C241 104
R143 5.1K
R144 5.1K
J11
CH340_P
CH340_N
+5V
U16
VCC
R232
RTS#
DTR#
DCD#
RI#
DSR#
CTS#
GND
TXD
RXD
V3
D+
D-
XI
XO
CH340C
C242 10uF
C243 104
UART_RXD
UART_TXD
Design Name
ZYNQ-Micro-SYS
Size
Page Name
A4
12_USB_OTG
Rev
1.2
Date:
Tuesday, October 12, 2021
Sheet
12
of
15



Design Name	ZYNQ-Micro-SYS	
Size	Page Name: A4 — 13_HDMI	Rev: 1.2
Date: Tuesday, October 12, 2021	Sheet 13 of 15	


5
4
3
2
1
D
C
B
A
XADC_VP
XADC_VN
R161 49R9
R162 49R9
C260
1nF
J14
Header_1*4
+3.3V
+1.8V
C239
C236 104
U15
TXS02612RTWR
VCCA
VCCB0
DATA0A
DATA1A
DATA2A
DATA3A
CLKA
CMDA
SEL
GND
GND
EPAD
DATA0B0
DATA1B0
DATA2B0
DATA3B0
CLKB0
CMDB0
VCCB1
DATA0B1
DATA1B1
DATA2B1
DATA3B1
CLKB1
CMDB1
SD_D0
SD_D1
SD_D2
SD_D3
SD_CLK
SD_CMD
R125 4.7K
R126 4.7K
R127 4.7K
R128 4.7K
R129 4.7K
R130 4.7K
C237 10uF
C238 104
J5
Micro SD
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
SD-CD
GND
R132 10K
R131 330R
PS_MIO10
SD_CD
MIO4 MIO5
BOOT CFG 1 2
JTAG ON ON
QSPI ON OFF
SD Card OFF OFF
J10
BOOT_CFG
R141 10K
R142 10K
R139 20K
R140 20K
BOOT_MODE0
BOOT_MODE1
QSPI_D2
QSPI_D3
J15
JATG_2*7P
R137 33R FPGA_TMS
R135 33R FPGA_TCK
R136 33R FPGA_TDO
R138 33R FPGA_TDI
FPGA_DONE
LED1
PL LED
B35_IO0
PL_LED
R3 330R
R0402
red
LED2
PS LED
PS_LED
R4 330R
red
FPGA DONE LED
LED3
R5 330R
R92 220R
green
POWER LED
LED4
R6 330R
bule
PS_MIO12
B35_IO25
PS_KEY
PL_KEY
SW1
SW2
KEY
R133 10K
R134 10K
Design Name
ZYNQ-Micro-SYS
Size
Page Name
A4
14_Device
Rev
1.2
Date:
Tuesday, October 12, 2021
Sheet
14
of
15



Design Name	ZYNQ-Micro-SYS	
Size	Page Name A4	15_GPIO Rev 1.2
Date: Tuesday, October 12, 2021	Sheet 15 of 15	